

Lesson 2: Integrating the equations of motion

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1. The acceleration of a drone is given as

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$$a = 16 - 3t^2$$

where a is in ft/s^2 and t is in seconds. The drone starts at $t=0$ with $v=0$ at $x=100$ feet. Determine (a) the time when the velocity is again zero, (b) the position when $t=6$ s , (c) the velocity when $t=6$ s and, (d) the total distance traveled by the drone from $t=0$ to $t=6$ s.

2. A car traveling at 50 mph hits a highway crash barrier. Upon impact the car decelerates at a rate

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$$a = -45\sqrt{x}$$

where a and x have dimensions of ft/s^2 and ft , respectively. How far will the car travel into the barrier before coming to rest?

Answer: _____

3. A shock absorber utilizes a piston with orifices that pushes through a fluid-filled cylinder. At high speeds the piston decelerates at a rate proportional to the velocity to the $2/3$ power, $a = -kv^{2/3}$.

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A testing rig starts at $t=0$ with $v=64$ in/s. The piston is observed to have a velocity $v=8$ in/s at time $t=3$ in. Determine (a) the velocity of the piston at $t=1.5$ s, (b) the time required for the shaft to come to rest, (c) the maximum travel of the piston.
